

Call or Fold? - Assignment 3

SciMathSoc

June 26, 2026

Instructions

- You are required to submit through github at [Call or Fold?](#)
- I won't entertain any submission on WhatsApp DMs.
- Ask Doubts regarding the assignment
- If you still have problems working with github, ask me.
- Make a folder for Assignment-3 to submit your colab notebook and first problem.
- **Deadline : July 1, 2026**
- Submitting this assignment is mandatory for ratification (every subsequent assignment, even the ones given by Chaitanya).

Question 1

OX Expected Score

You're playing a game called *OX!*. There are n clicks in a game. For each click there are two possible outcomes:

- Correct click, denoted by **O**,
- Bad click, denoted by **X**.

Thus, a play can be represented as a sequence of n characters consisting of **O** and **X**.

The score of a play is calculated as follows:

- Consider every maximal consecutive block of **O**'s.
- If a block has length L , it contributes L^2 points to the score.
- The total score is the sum of contributions of all such blocks.

For example, the play

OOXOOOXXOO

contains three maximal consecutive **O**-blocks:

OO, OOO, OO.

Therefore, the score is

$$2^2 + 3^2 + 2^2 = 17.$$

If there are no correct clicks, the score is 0.

You are given the probability p_i that the i -th click ($1 \leq i \leq n$) is correct. In other words,

$$\Pr(\text{click } i \text{ is } 0) = p_i,$$

and

$$\Pr(\text{click } i \text{ is } \mathbf{X}) = 1 - p_i.$$

The outcomes of different clicks are independent.

Task: Compute the expected score of the play.

NOTE: For this task, either submit a proof-based solution in L^AT_EX or handwritten format. Also, to flex your programming skills, you can alternatively write code for the same with specifying appropriate core logic and implementation behind the problem.

To validate your solution, check out this problem [Problem - B - Codeforces](#) (do not look at the editorial and copy the solution, I'll deratify you straightaway) and submit it (don't judge the problem by its rating and tags, it can be done without knowledge of DP). The person who submits this problem will receive a guaranteed treat from me (up to Rs. 100). Take hints from Me or Chaitanya if needed.

Question 2

Use this colab link for the question, most of the things have been done for you. Y'all just need to muster up your game theory preliminaries and use basic statements and operators.

Link: [Assignment-3-Call or Fold? - Colab](#)